

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Lesson 4: Classification and Evaluation Metrics

## Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

30 October 2026

# Before we start

- Exercise 3, which we discuss now
- Ridge, Lasso, and the choice of the penalty strength

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1, and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**
- Answer with a probability: $0 \leq P(y = 1 \mid x) \leq 1$
- $y = 1$ if the drive failed, $y = 0$ if not; $x$ holds its six counters

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each: Self-Monitoring, Analysis and Reporting Technology
- Reallocated sectors, spin retries, read and seek error rates, temperature, hours in service
- Label: did it fail within thirty days?
- `train_test_split(..., stratify=y)`: 6,000 to train, 2,000 to test

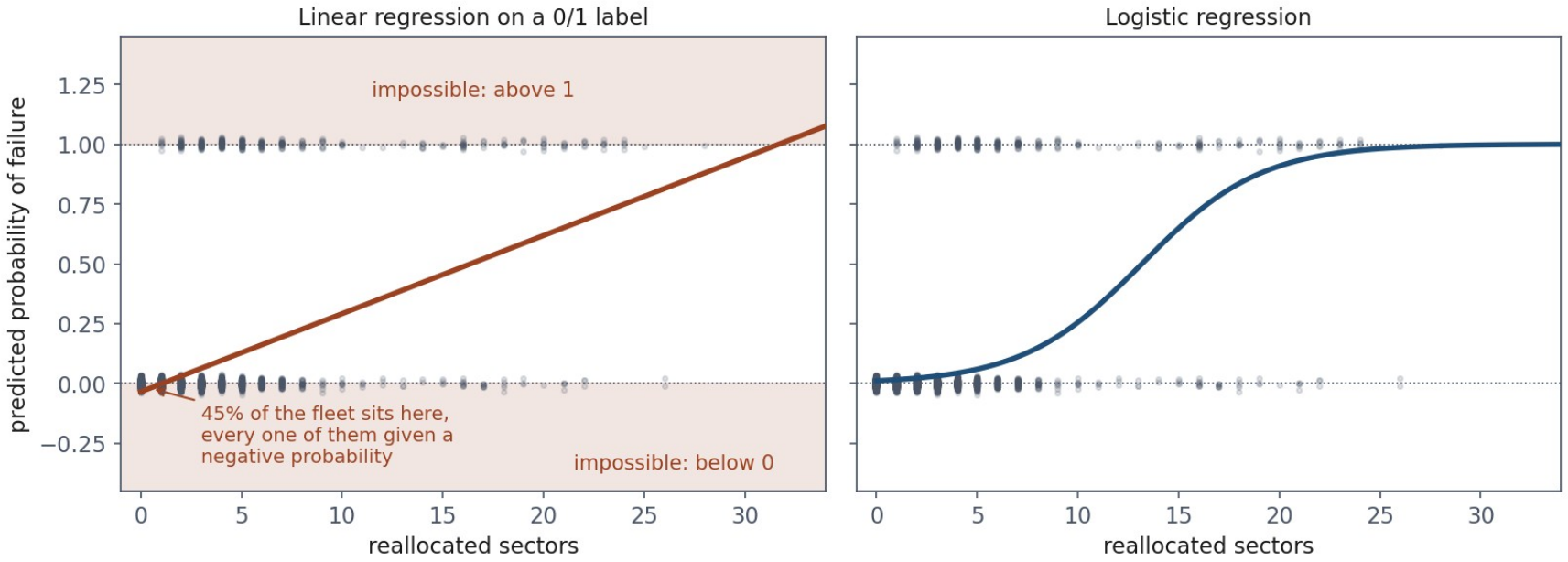
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# 306

- **306 of the 8,000 drives failed.** 3.8% of the fleet
- Which means 96.2% did not

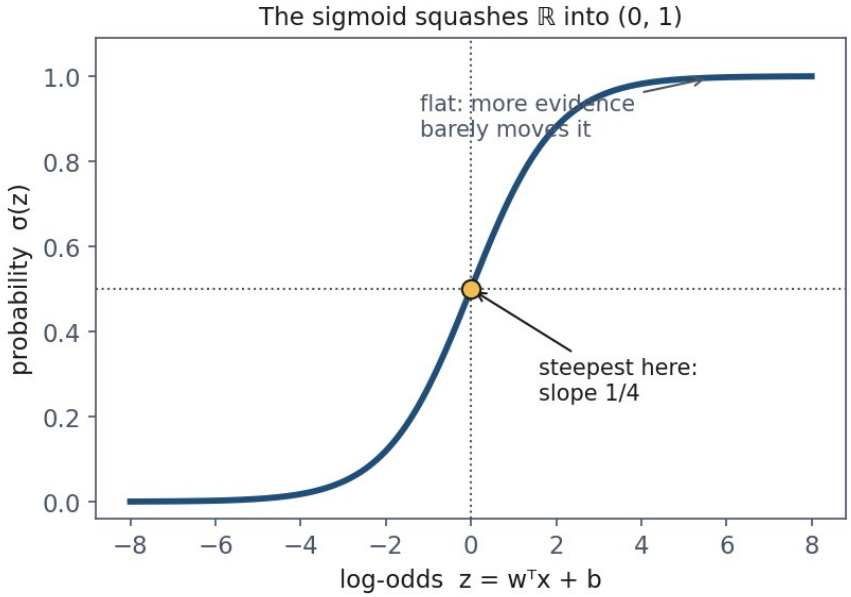
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

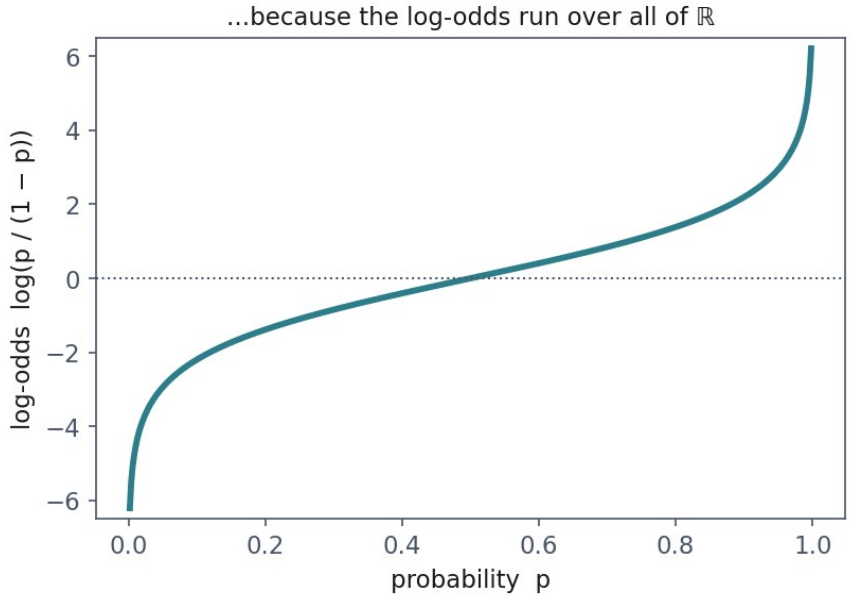
# What goes wrong if we just fit a line




UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# The sigmoid




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Why the log-odds

- A line gives any number; a probability, only [0, 1]
- Odds, $p \div (1 - p)$: no ceiling, from 0 to $\infty$
- Log of the odds: no floor either, $-\infty$ to $+\infty$
- $p = 0.01, 0.5, 0.99 \rightarrow$ log-odds $-4.60, 0, +4.60$
- Bonus: coefficients multiply the odds (slide 12)

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Log-odds of p = P(y = 1 | x): a line

$$\log \frac{p}{1-p} = w^\top x + b \quad \Longrightarrow \quad p = \frac{1}{1 + e^{-(w^\top x + b)}}$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# What the coefficients mean

- The model is linear in the log-odds, log(p ÷ (1 − p)), so `exp(coefficient)` multiplies the **odds**, p ÷ (1 − p)
- Features standardised, so one unit = one standard deviation
- A coefficient of 1.80 means **six times the odds** of failing
- In code: `make_pipeline(StandardScaler(), LogisticRegression())`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The truth, and what we recovered

| Feature | True | Fitted | Odds × |
|---|---|---|---|
| reallocated_sectors | 1.80 | 1.66 | 5.28 |
| spin_retry_count | 1.15 | 1.17 | 3.22 |
| read_error_rate | 0.85 | 0.87 | 2.39 |
| temperature_c | 0.45 | 0.47 | 1.59 |
| power_on_hours | 0.35 | 0.38 | 1.46 |
| seek_error_rate | **0.00** | **0.02** | **1.03** |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The intercept is the average drive’s chance

- Fitted intercept: **−6.09**
- σ(-6.09) = 0.0023: a 0.2% chance for a drive average on every counter
- The fleet fails at 3.8%: the failures come from the few drives far above average

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Fitting means minimising something, but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid’s output?

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Step 1: each drive gets a p

| Drive | What happened | y | Setting 1: $p = \sigma(w^T x + b)$, the P(fail) |
|---|---|---|---|
| A | failed | 1 | 0.8 |
| B | stayed healthy | 0 | 0.1 |
| C | stayed healthy | 0 | 0.2 |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Step 2: how likely was the outcome?

| Drive | y | p | Probability of what happened: p if y = 1, 1 - p if y = 0 |
|---|---|---|---|
| A | 1 | 0.8 | 0.8 |
| B | 0 | 0.1 | $1 - 0.1 = 0.9$ |
| C | 0 | 0.2 | $1 - 0.2 = 0.8$ |
| All three: multiply | | | $0.8 \times 0.9 \times 0.8 =$ **0.576** |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Step 2 as formulas: the likelihood

$$P(y \mid x) = p^{\,y}\,(1-p)^{1-y}$$

$$\mathcal{L}(w,b) = \prod_{i=1}^{m} \left(p^{(i)}\right)^{y^{(i)}} \left(1-p^{(i)}\right)^{1-y^{(i)}}$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Step 3: two settings, the likelier wins

| Drive | y | Setting 1: p | Setting 1: probability of what happened | Setting 2: p | Setting 2: probability of what happened |
|---|---|---|---|---|---|
| A | 1 | 0.8 | 0.8 | 0.3 | 0.3 |
| B | 0 | 0.1 | 0.9 | 0.6 | 0.4 |
| C | 0 | 0.2 | 0.8 | 0.5 | 0.5 |
| Likelihood: the product | | | **0.576** | | **0.060** |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Step 4: minus the log, a cost

| Drive | y | Setting 1: −log of probability of what happened | Setting 2: −log of probability of what happened |
|---|---|---|---|
| A | 1 | −log 0.8 = 0.22 | −log 0.3 = 1.20 |
| B | 0 | −log 0.9 = 0.11 | −log 0.4 = 0.92 |
| C | 0 | −log 0.8 = 0.22 | −log 0.5 = 0.69 |
| Sum = −log of the likelihood | | **0.55** | **2.81** |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Cross-entropy, also called log loss

$$L = -[\, y \log p + (1-y) \log(1-p)]$$

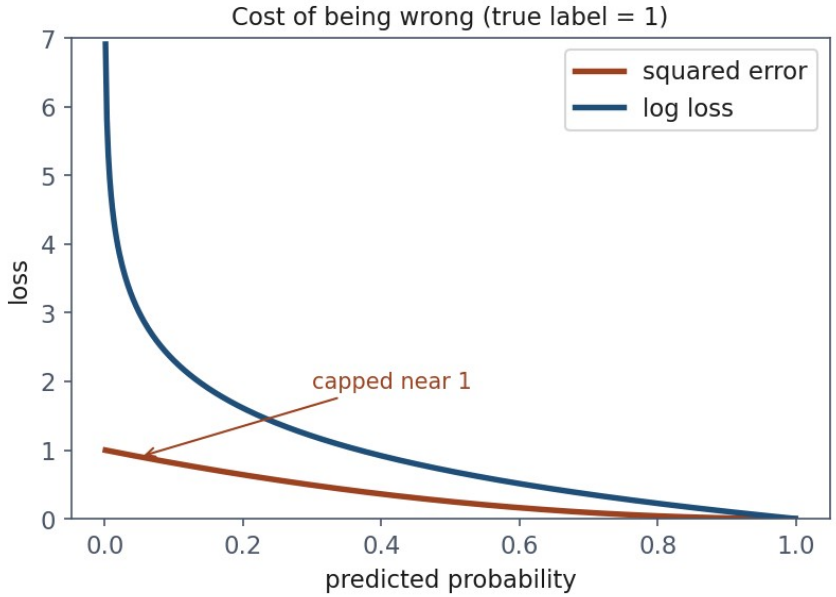
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

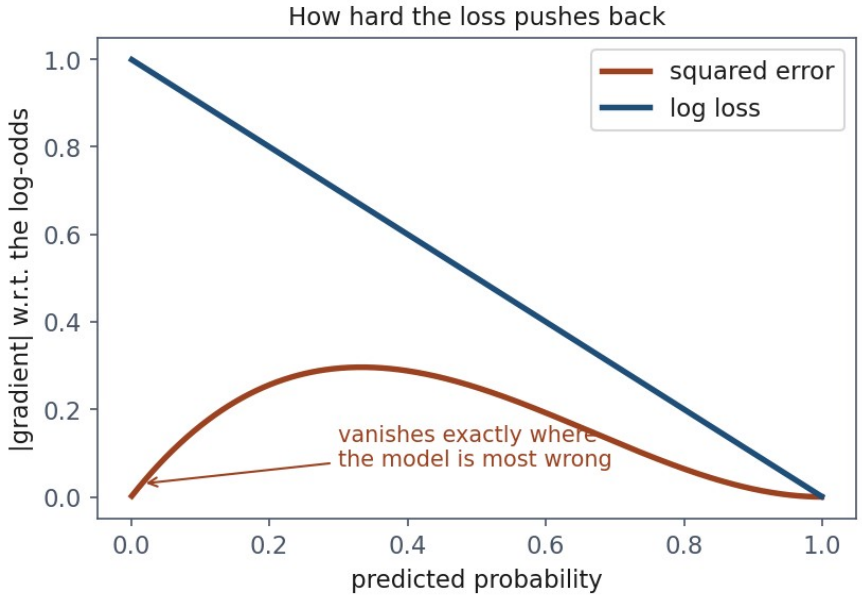
# Lesson 3’s gradient, $\hat{y} = \sigma(w^T x + b)$

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# So why not squared error?




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not

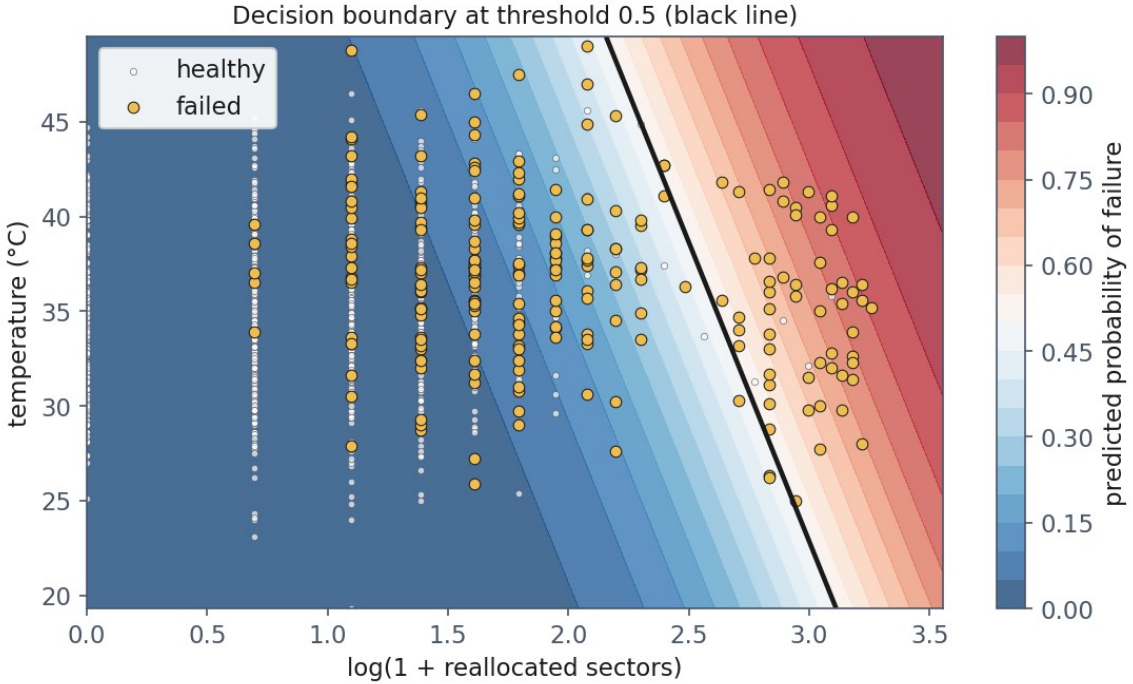
UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Notebook 1, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| a line on a 0/1 label, 7 | `LinearRegression` → 3,606 negative probabilities |
| odds, and why not squared error, 8-24 | NumPy only: the sigmoid, the two losses, their gradients |
| gradient descent, 22 | `train_test_split`, `StandardScaler`, fifteen lines of NumPy → log loss 0.6931 to 0.0692 |
| the decision boundary, 26 | `make_pipeline(StandardScaler(), LogisticRegression())` on two features |
| the truth, recovered, 13-14 | `LogisticRegression`, `.coef_`, `.intercept_` → seek_error_rate 0.02, intercept −6.09 |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The decision boundary




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Break

- Ten minutes

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Two models

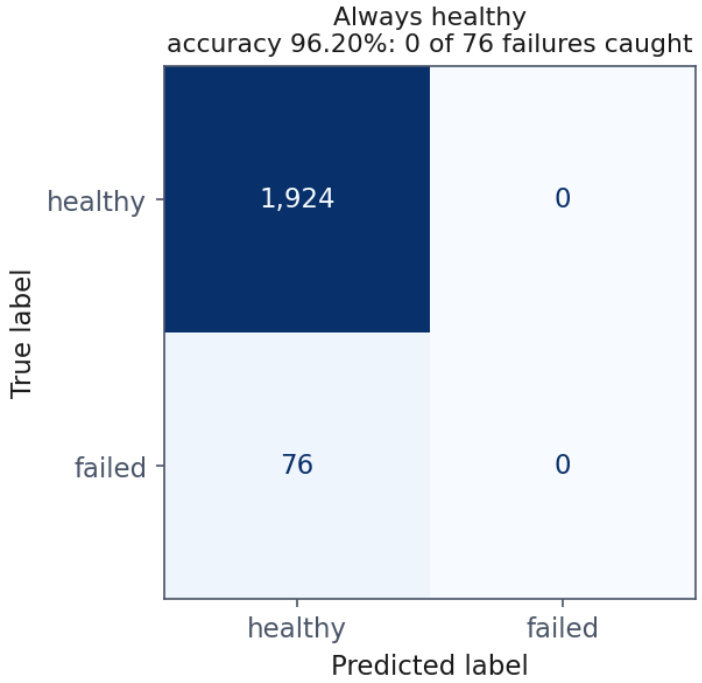
| Model | Accuracy | Failures caught |
|---|---|---|
| Predict “healthy” for everything | **96.20%** | **0 of 76** |
| Logistic regression, threshold 0.5 | 97.70% | 43 of 76 |

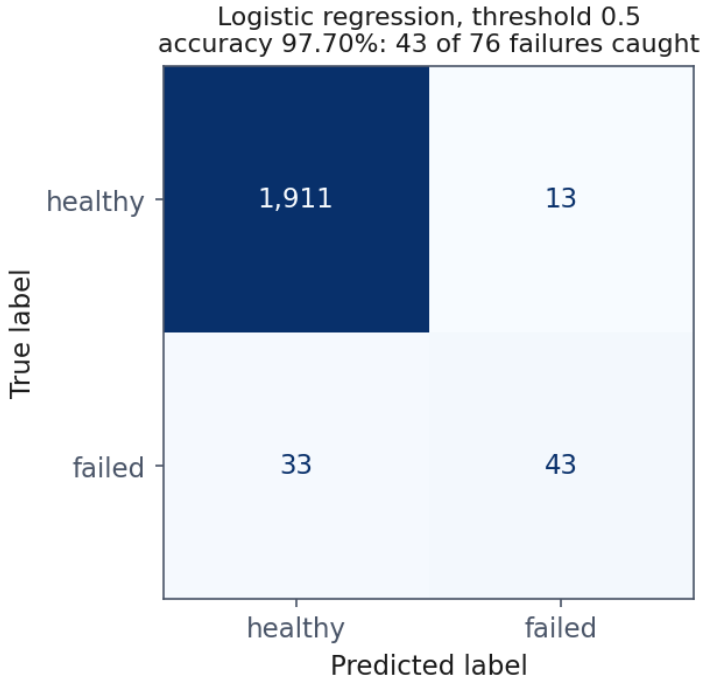
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem that is exactly what you want
- The interesting class is almost always the **rare** one

# Four numbers instead of one




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The four outcomes, and what they cost

- **False positive**: we replace a healthy drive. A technician’s hour, €140
- **False negative**: a drive dies in service. Callout, rebuild, risk, €2,600

# Precision and recall: look at the denominator

Precision = TP / (TP + FP)
"of the alarms we raised"

| | predicted healthy | predicted failing |
|---|---|---|
| actually healthy | TN<br>1911 | **FP**<br>13 |
| actually failed | FN<br>33 | **TP**<br>43 |

Recall = TP / (TP + FN)
"of the drives that failed"

| | predicted healthy | predicted failing |
|---|---|---|
| actually healthy | TN<br>1911 | FP<br>13 |
| actually failed | **FN**<br>33 | **TP**<br>43 |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Our model, in those terms

- True and false positives and negatives: TP, FP, FN, TN
- Precision = TP ÷ (TP + FP) = 43 ÷ 56 = **0.768**: of the drives flagged
- Recall = TP ÷ (TP + FN) = 43 ÷ 76 = **0.566**: of the drives that failed
- Specificity = TN ÷ (TN + FP) = 1,911 ÷ 1,924 = 0.993
- In code: `precision_score`, `recall_score`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# F1: why the harmonic mean

| Model | Precision | Recall | Arithmetic mean | Harmonic, F1 = 2PR ÷ (P + R) |
|---|---|---|---|---|
| Flag every drive | 0.038 | 1.000 | 0.519 | **0.073** |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Read the minority class, not the average

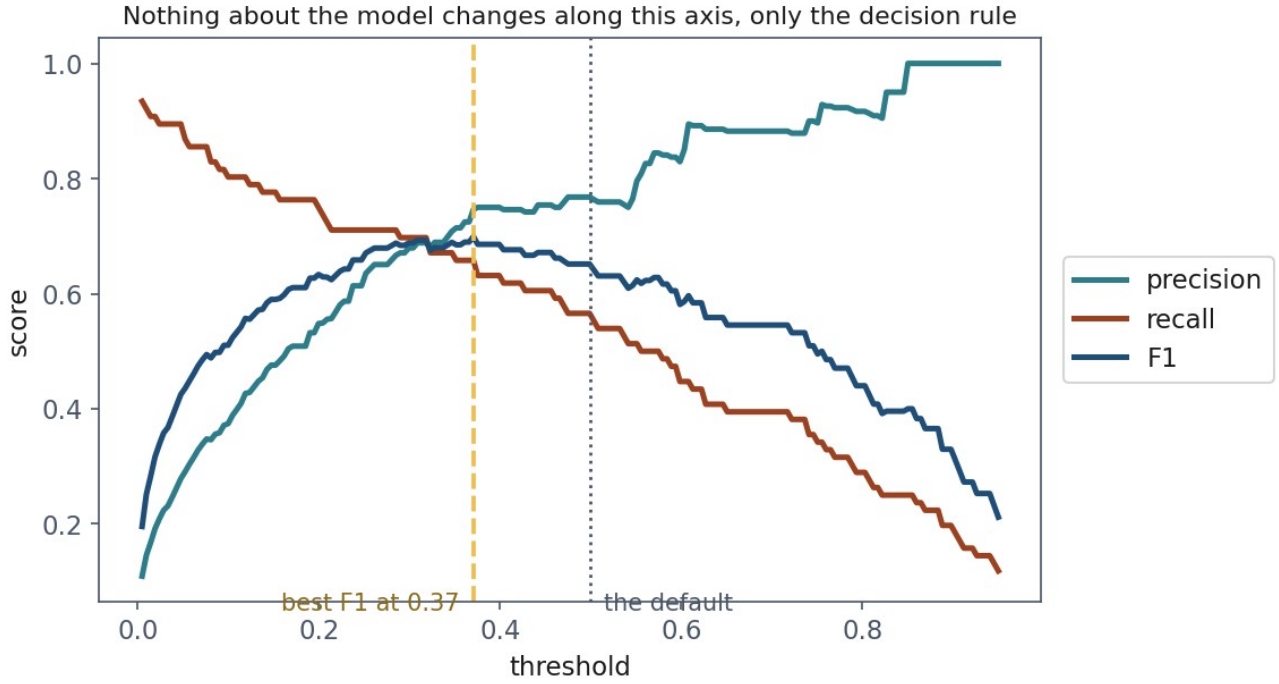
- `weighted avg` F1: **0.975** (dominated by the healthy class)
- `macro avg` F1: **0.820** (both classes get an equal say)
- In code: `classification_report(y_test, y_pred)` prints both rows

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Notebook 2, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| the number that should worry you, 28 | `accuracy_score` → 96.20% for “always healthy” |
| the confusion matrix, 30 | counts by hand, then `confusion_matrix`, `ConfusionMatrixDisplay` → TP 43 |
| precision and recall, 32-35 | `precision_score`, `recall_score`, `f1_score`, `classification_report` |
| the threshold sweep, 37-38 | `predict_proba`, then the metrics at each threshold |
| the precision-recall curve, 39 | `precision_recall_curve`, `average_precision_score` → 0.717 against 0.038 |

# The threshold is a choice nobody made



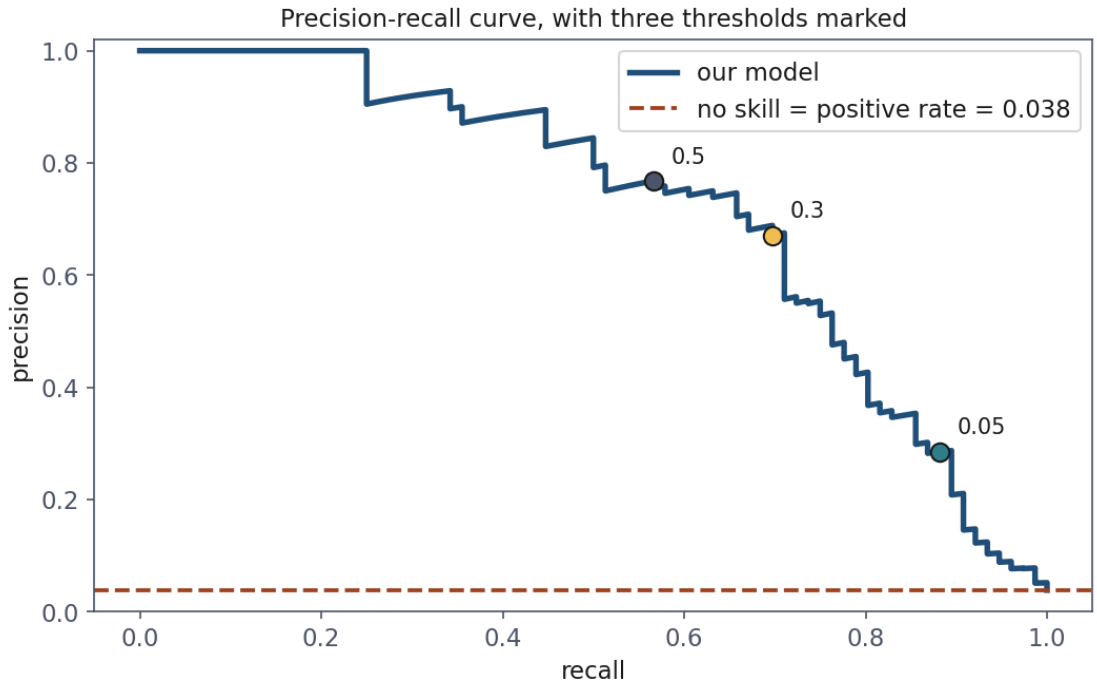

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# What the sweep actually costs

| Threshold | TP | FP | FN | Precision | Recall | Accuracy |
|---|---|---|---|---|---|---|
| 0.02 | 69 | 284 | 7 | 0.195 | 0.908 | 0.855 |
| 0.10 | 61 | 102 | 15 | 0.374 | 0.803 | 0.942 |
| 0.30 | 53 | 25 | 23 | 0.679 | 0.697 | 0.976 |
| 0.50 | 43 | 13 | 33 | 0.768 | 0.566 | **0.977** |
| 0.90 | 14 | 0 | 62 | 1.000 | 0.184 | 0.969 |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The precision-recall curve



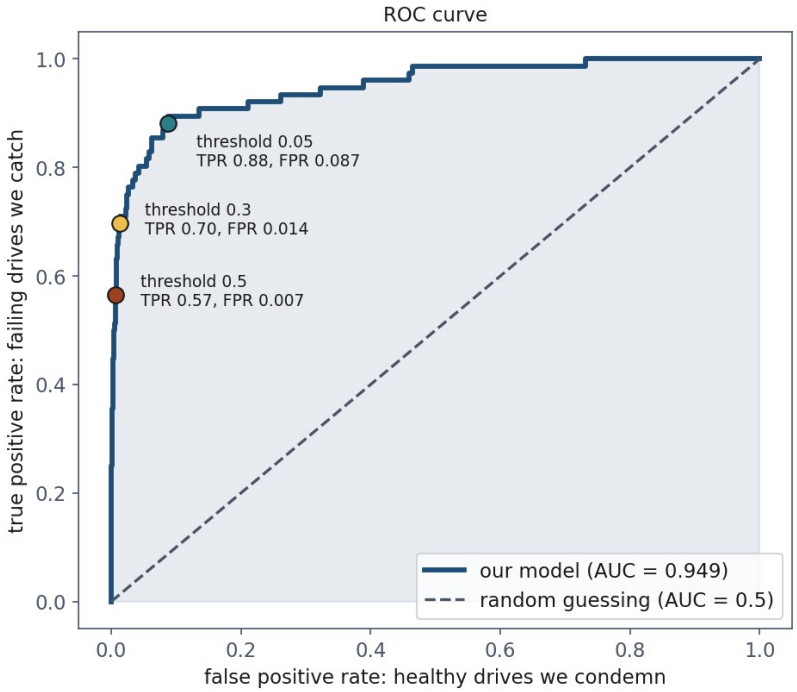

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Comparing models without choosing a threshold

- We keep evaluating at one threshold, but which?
- What if we score the whole **ranking** instead?

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The ROC curve




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# What the area means

- **Area under the curve (AUC) = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949
- In code: `roc_auc_score(y_test, scores)`

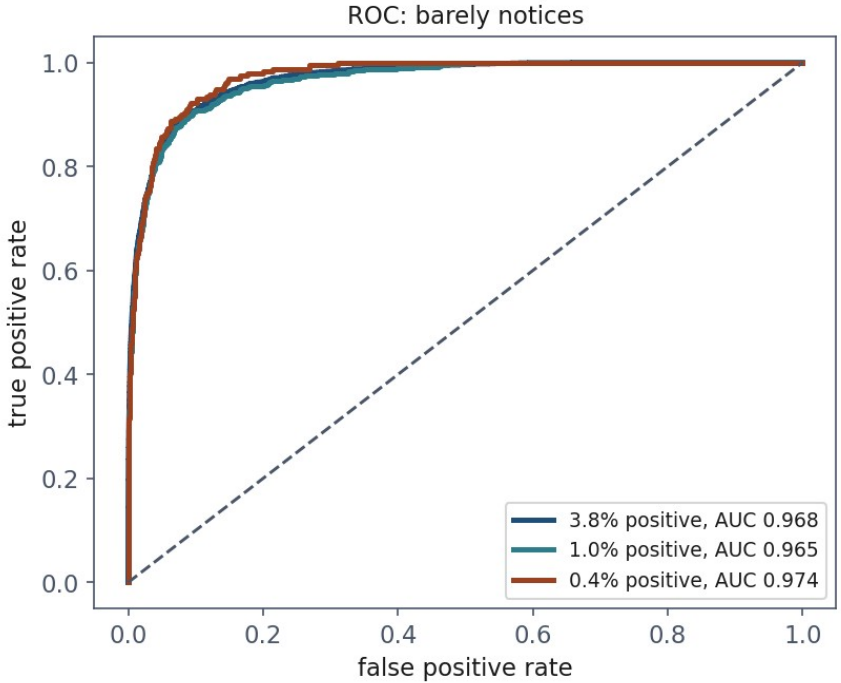
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

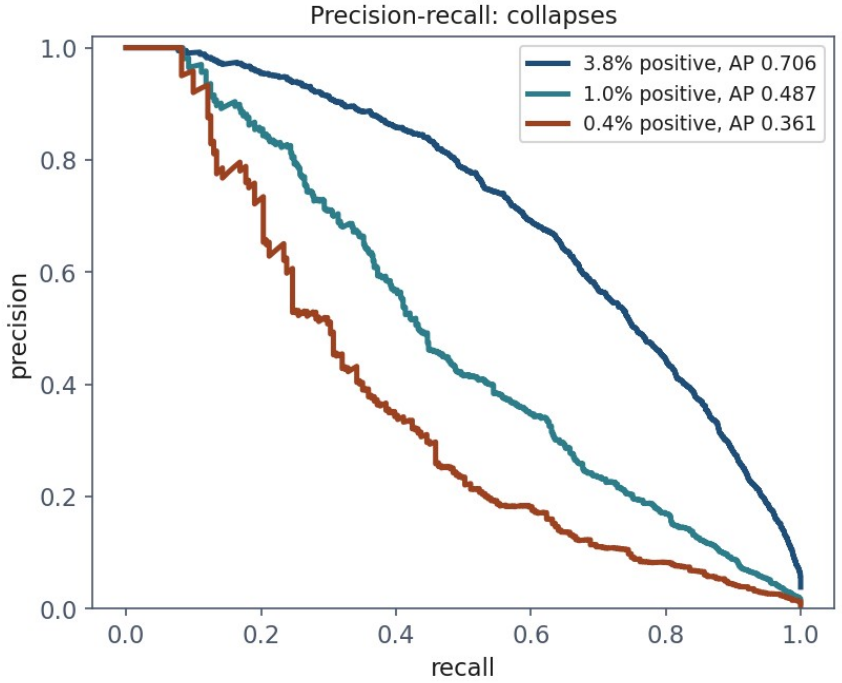
# What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**; every decision changes

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# Where ROC misleads




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Why the two disagree

- False positive rate (FPR) divides by **all healthy drives**: an enormous, growing denominator
- Precision divides by **the alarms raised**, where the false ones dominate

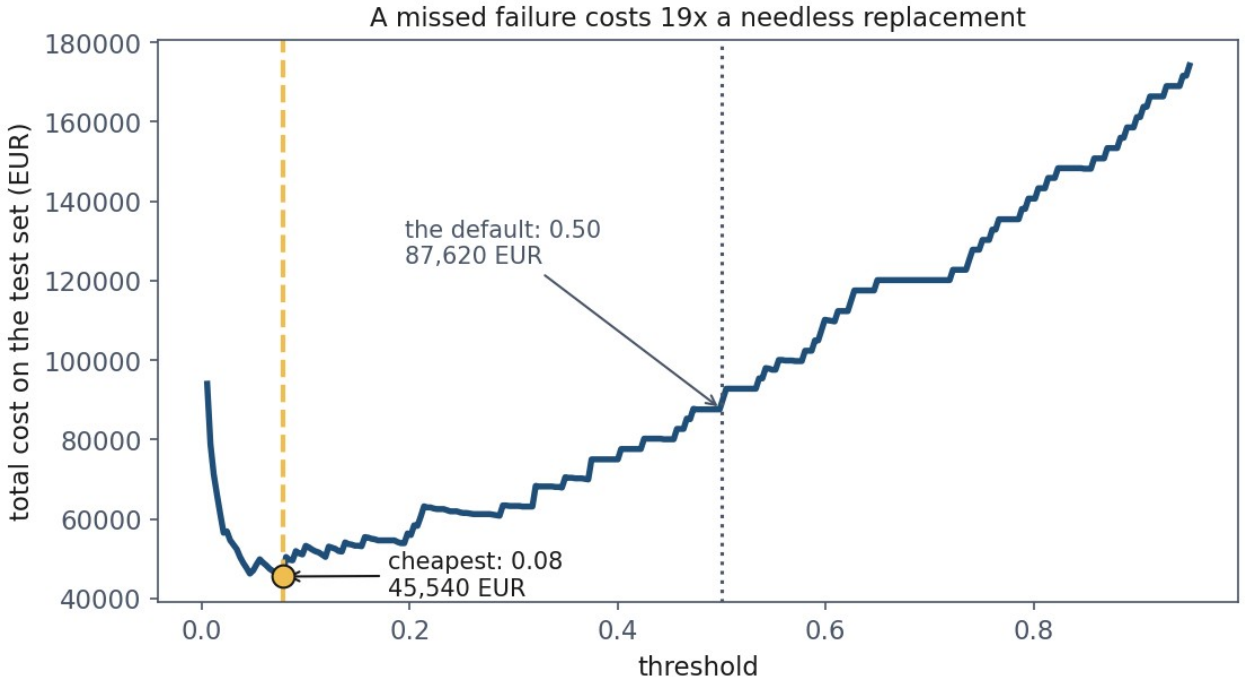
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# The rule

- On an imbalanced problem, report the **precision-recall curve alongside AUC**
- Never instead of it, and never AUC alone
- Its area is the **average precision**, `average_precision_score`: 0.717 here, against a floor of 0.038

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Choosing the threshold from money




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# And it has a closed form

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{140 + 2600} = 0.051$$

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade

# Class weights: the same lever, pulled earlier

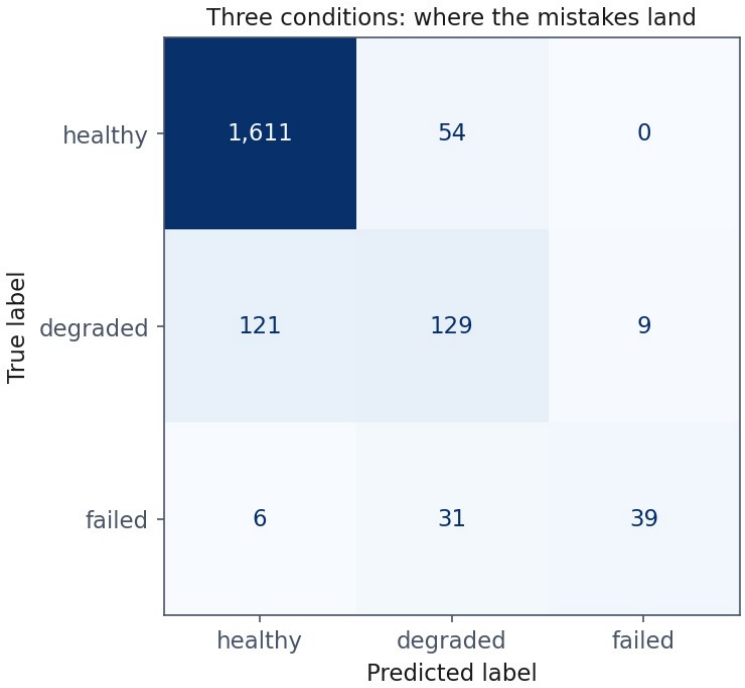
| Model | Recall | Precision | AUC | Cost |
|---|---|---|---|---|
| Plain, threshold 0.50 | 0.566 | 0.768 | 0.949 | 87,620 |
| Plain, threshold 0.08 | 0.855 | 0.349 | 0.949 | 45,540 |
| Balanced class weights, 0.50 | 0.895 | 0.249 | **0.950** | 49,500 |

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Notebook 3, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| the ROC curve by hand, 41 | `roc_curve`, `roc_auc_score` → 0.9493 |
| what AUC means, 42 | 200,000 random (failing, healthy) pairs → 0.9488 |
| the rare-positive trap, 44-45 | `average_precision_score` beside `roc_auc_score` → AP 0.706 to 0.361, AUC unmoved |
| the threshold from costs, 47-49 | `confusion_matrix` at each threshold → 45,540 EUR at 0.08, against 87,620 at 0.50 |
| class weights, 50 | `LogisticRegression` with balanced class weights → AUC 0.950 |
| three classes, 52-53 | `classification_report` → macro F1 0.707, weighted 0.883 |

# More than two classes




UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Macro against weighted, one last time

- macro avg F1: **0.707**
- weighted avg F1: **0.883**

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

# What to take away

- **96.20% accuracy, nothing caught.** Accuracy measures the majority class
- Precision is about the **alarms**, recall is about the **failures**
- The **threshold is a decision**, and it has a formula
- **AUC measures ranking** and hides imbalance

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

# Homework

- **Exercise 4**, discussed at the start of **Friday 6 November**
- Build a classifier, choose a threshold from a stated cost ratio, defend it

UNIVERSITÀ DEGLI STUDI DELL'AQUILA